

ChargeCascade: A Tumbling-Mill Charge-Motion and Power Workbench, and the Disagreement of Standard Power Models

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 Not peer reviewed. Source code, data and computational records: github.com/fsantibanezleal/CAOS_ChargeCascade

Abstract. The power a tumbling mill (SAG, ball or rod) draws is the single largest electricity cost in a concentrator, and predicting it, and the charge motion behind it, is a classic problem with more than one accepted answer. ChargeCascade is an interactive workbench that computes tumbling-mill charge motion and power live in the browser from geometry, fill, ball size and the fraction of critical speed, and it shows where the accepted models disagree. The charge motion follows Davis (1919) kinematics: a per-shell departure angle traces the shoulder, toe and cataract fan across the cascading, cataracting and centrifuging regimes, with the centrifuging onset landing exactly at the critical speed. The power is computed by two standard closed-form models, Hogg-Fuerstenau (with a stated effective-lift-arm calibration) and Morrell, and the central finding is that these two textbook models disagree by about 23% (median, up to 29%) across ten cases, with a third estimate from a baked discrete-element simulation (5000–10000 particles) landing around them rather than adjudicating between them. This spread, a cross-method uncertainty of order twenty percent between standard models, is displayed with all three estimates side by side. A learned surrogate reproduces the closed-form power to 5.2% for instant what-ifs, guarded by an out-of-distribution detector that flags settings outside its training envelope. A single dependency-free engine computes the physics; every number and figure can be regenerated from the released artifacts.

Index Terms. Tumbling mill, SAG mill, ball mill, charge motion, mill power, Davis kinematics, Hogg-Fuerstenau, Morrell, discrete element method, surrogate model, comminution.

I. INTRODUCTION

A tumbling mill grinds ore by lifting a charge of balls and rock on a rotating shell and letting it cascade and cataract back down, and the power it draws to do so is the dominant energy cost of a concentrator. Predicting that power, and the charge motion that produces it, has a century of history and, importantly, more than one accepted model. The charge kinematics trace back to Davis [1]: at a fraction $\phi_c = N/N_c$ of the critical speed $N_c = 42.3/\sqrt{D-d}$, a ball on a shell of radius r leaves the shell at a departure angle given by $\cos \alpha = \phi_c^2 r/R$ and follows a ballistic arc, and summing

over shells gives the shoulder, the toe and the cataract fan. The power has at least two standard closed-form answers, the Hogg-Fuerstenau torque-arm model [2] and the Morrell model [3], and a higher-fidelity but expensive answer from the discrete element method [4], [5], [6]. These do not all agree, and a single reported value hides the spread between them.

ChargeCascade computes the charge motion and the power from the cited models, live and in the browser, and shows the reader where the models disagree. It runs Davis kinematics for the charge shape and regime, both the Hogg-Fuerstenau and the Morrell closed-form power models, and a set of baked discrete-element traces as a third estimate, and it puts them side by side. This report describes the workbench and its central finding: the two standard closed-form power models disagree by about a fifth, a cross-method uncertainty that the workbench displays instead of reporting one of the two values. A learned surrogate accelerates the closed-form engine, and an out-of-distribution guard flags extrapolation.

II. CHARGE MOTION: DAVIS KINEMATICS

The charge shape and regime follow from Davis kinematics evaluated over nine radial shells (Fig. 1). As the fraction of critical speed rises, the charge is carried higher before it departs, so the shoulder angle rises and the toe advances, and the motion passes through the literature regime bands: slumping and cascading at low speed (here $\phi_c \approx 0.55$ – 0.62 , shoulder near 70°), cataracting at intermediate speed ($\phi_c \approx 0.75$ – 0.78 , the ballistic fan that does the impact grinding in a SAG mill), and centrifuging at the critical speed. The centrifuging onset is exact: below $\phi_c = 1$ no shell centrifuges, and at $\phi_c = 1$ the outer shell pins to the wall, the shoulder collapses to zero and the toe to 180° (Fig. 1b). Because the critical speed and ϕ_c are derived from geometry, not fitted, the regime map is a property of the kinematics, and it reproduces the textbook picture, which is the check that licenses using the workbench to reason about charge motion.

III. POWER: WHERE THE STANDARD MODELS DISAGREE

The power is where the comparison between models matters most (Fig. 2, Table I). Two standard closed-form models are

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Interactive workbench and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_ChargeCascade; the workbench is available at <https://chargetascade.fasl-work.com>.

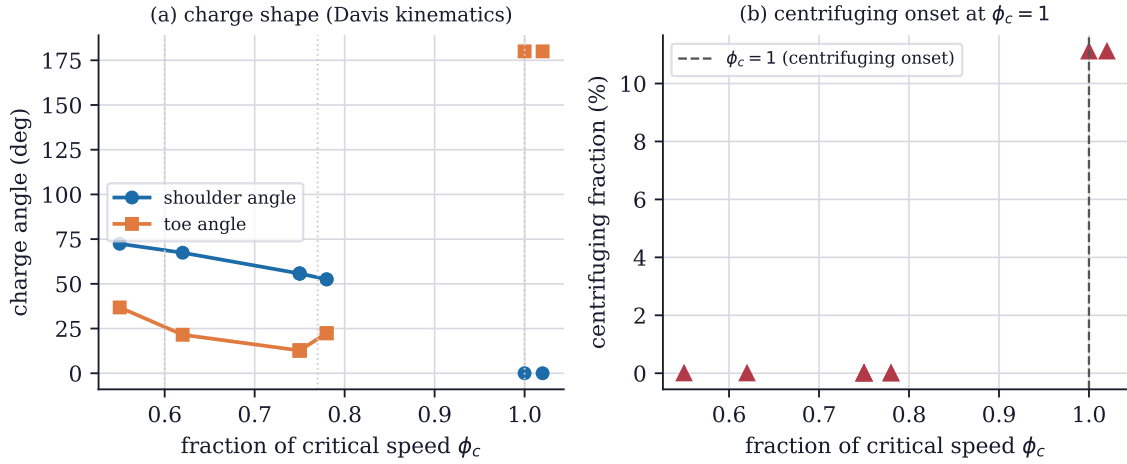


Fig. 1. Charge motion by Davis kinematics, across ten cases. (a) The shoulder and toe angles against the fraction of critical speed ϕ_c : as ϕ_c rises the charge lifts higher (the shoulder rises, the toe advances) through the cascading and cataracting regimes, and at the critical speed the charge centrifuges (the shoulder collapses to zero and the toe to 180°, the charge pinned to the shell). (b) The centrifuging fraction is zero below the critical speed and turns on exactly at $\phi_c = 1$, the exact onset Davis kinematics predicts.

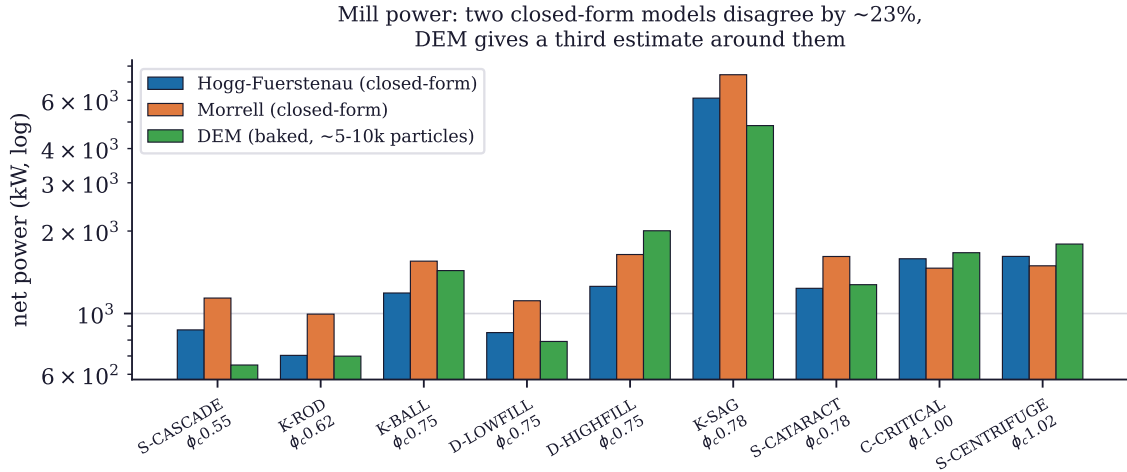


Fig. 2. Net mill power per case, on a log axis, from two closed-form models and from a baked discrete-element simulation. The Hogg-Fuerstenau and Morrell closed-form models *disagree* by *about* 23% (median; up to 29%): Morrell is consistently higher at cascading and cataracting speeds, and the two cross near the critical speed. The discrete-element estimate (5000–10000 particles) lands *around* them, sometimes below both, sometimes above, rather than settling the disagreement. This spread is the cross-method uncertainty of mill-power prediction, and the workbench shows all three rather than reporting one as fact.

computed: Hogg-Fuerstenau [2], a torque-arm model with an effective-lift-arm calibration ($C_{\text{arm}} = 0.80$, stated in the interface), and Morrell [3]. They do not agree. Across the ten cases the two disagree by a median of 23% and by as much as 29%: Morrell draws consistently higher at the cascading and cataracting speeds that mills actually run at (for example 1139 kW against 872 kW on a cascading case), and the two only cross near the critical speed. A third estimate, from a baked discrete-element simulation of five to ten thousand particles, lands *around* the two closed-form values, below both on some cases and above on others, rather than adjudicating between them. The spread is a property of the models, not of the workbench: on these cases the two standard models differ by 18 to 29 percent below the critical speed, so a single power reported to four significant figures would assert a precision

that the models do not support. ChargeCascade shows the two closed-form models and the discrete-element estimate together, with the Hogg-Fuerstenau calibration stated, so a user sees the spread and can reason about it.

IV. THE LEARNED SURROGATE AND ITS GUARD

For instant what-if exploration the workbench runs a learned surrogate of the closed-form engine: a small multilayer perceptron (six inputs, two hidden layers of sixty-four units, mapping to power and centrifuging fraction) trained on the engine's outputs. It reproduces the closed-form power to a mean relative error of 5.2% (with the centrifuging fraction exactly, since it is a kinematic threshold), fast enough to update as a user drags a control. Crucially, because a surrogate is only reliable inside the envelope it was trained on, a companion out-of-distribution

TABLE I

NET POWER BY THE TWO CLOSED-FORM MODELS AND THE BAKED DEM FOR SIX REPRESENTATIVE CASES OF THE TEN; THE EMPTY-MILL CASE C-EMPTY, WHOSE CLOSED-FORM POWER IS ZERO, IS OMITTED. BELOW THE CRITICAL SPEED THE CLOSED-FORM MODELS DISAGREE BY 18–29% OF THE LARGER VALUE, AND THEY CROSS NEAR IT (8% AT C-CRITICAL); DEM LANDS AROUND THEM.

Case	ϕ_c	Hogg-F. (kW)	Morrell (kW)	DEM (kW)
S-CASCADE	0.55	872	1139	649
K-ROD	0.62	704	995	699
K-BALL	0.75	1188	1553	1435
D-HIGHFILL	0.75	1256	1642	2005
K-SAG	0.78	6111	7440	4852
C-CRITICAL	1.00	1584	1464	1667

detector flags settings outside that envelope, so a user who pushes the sliders to an untrained corner is warned that the surrogate is extrapolating rather than being silently misled. A learned DEM surrogate, a graph-network simulator trained to roll the particle dynamics forward, is under development; it is not validated, and no result reported here uses it. The surrogate accelerates the engine; the underlying model disagreement remains displayed beside it.

V. DISCUSSION AND SCOPE

The workbench’s contribution is as much epistemic as computational: it computes tumbling-mill charge motion and power from the cited models and, by showing the Hogg-Fuerstenau and Morrell models and the discrete-element estimate together, it makes the cross-method uncertainty of mill-power prediction visible. The Hogg-Fuerstenau effective-lift-arm calibration is a stated constant, not a fit to a specific mill, and the interface shows it; the Morrell model is used as published. The cases are physically reasonable synthetic configurations, not a specific plant, so the numbers are the models’ behaviour, not a prediction for a given mill. The baked discrete-element traces are a higher-fidelity but still-modelled estimate at laboratory particle counts, not a measured power, and the graph-network DEM surrogate is still under development. The value is a reproducible workbench that presents mill-power prediction together with its cross-method spread. The interface states the same limits on its methodology pages.

VI. CONCLUSION

ChargeCascade computes tumbling-mill charge motion by Davis kinematics, reproducing the cascading-cataracting-centrifuging progression with an exact centrifuging onset at the critical speed, and computes the power by two standard closed-form models plus a baked discrete-element estimate. Its central finding is that the two textbook power models disagree by about a fifth, and it displays that spread with the third estimate around it, rather than reporting one number as fact, because a fifth is the cross-method uncertainty between these models. A learned surrogate reproduces the closed-form power to 5.2% with an out-of-distribution guard. The contribution is a reproducible workbench that displays the model disagreement and states its one calibration constant.

APPENDIX A REPRODUCTION

The repository artifacts under `data/derived/` carry the ten cases’ engine outputs (`case-results.json`: ϕ_c , regime, shoulder and toe angles, centrifuging fraction, and both the Hogg-Fuerstenau and Morrell power per case), the learned-surrogate metrics (`cc-learned.json`) and the baked DEM summaries (`data/dem/`). The figure scripts under `manuscripts/mill-charge-power/figures/` read those artifacts. The same dependency-free physics engine serves the interactive workbench, the experiment tables and the offline bake, so re-running reproduces Table I and Figs. 1–2.

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